

Roll No.

CENTRAL BOARD OF BACKEND DEVELOPMENT

PRACTICE EXAMINATION 2026

SUBJECT: DATABASE MANAGEMENT (SQL SERVER)

Time Allowed: 3 Hours

Maximum Marks: 100

General Instructions:

- This question paper contains 4 sections: A, B, C, and D.
- **Section A:** Objective/Very Short Answer Type Questions (20 Marks). Point-to-point answers only.
- **Section B:** Short Answer Type Questions (20 Marks). Brief syntax or 1-2 lines of explanation.
- **Section C:** Practical Query Writing (40 Marks). Strict syntax required.
- **Section D:** Case Study Based Question (20 Marks). Read the scenario carefully before writing code.
- Assume all tables are part of a SQL Server (T-SQL) environment.

SECTION A

(Very Short Answer Type Questions - 4 Marks Each)

- | | |
|--|----------|
| 1. Name the specific DDL command used to delete a table entirely, and the DDL command used to empty a table while keeping its structure. | 4 |
| 2. Can a single database table have multiple UNIQUE constraints? Answer Yes or No with a one-line reason. | 4 |
| 3. State the primary difference between UNION and UNION ALL regarding duplicate records. | 4 |
| 4. Which clause must strictly accompany an aggregate function if you want to filter grouped data? | 4 |
| 5. Write the syntax snippet to rename a column in an existing table. | 4 |

SECTION B

(Short Answer Type Questions - 5 Marks Each)

6. Differentiate between RANK () and DENSE_RANK (). Give a one-sentence practical example. *(Note: Important for interviews)* 5
7. Write an ALTER TABLE query to add a DEFAULT constraint setting the City column to 'Surat'. 5
8. Write a SQL query to fetch the current system date and time, and extract only the current Year from it. 5
9. Write a query using string functions to find employees whose FullName starts with 'M' and is exactly 5 characters long. 5

SECTION C

(Practical Query Writing - 10 Marks Each)

Reference Tables for Section C:

Employee (EmpID INT, FullName VARCHAR(50), Salary INT, DeptID INT, City VARCHAR(30))

Department (DeptID INT, DeptName VARCHAR(50))

10. Using a Subquery, write a query to find the 3rd highest salary from the Employee table. (Do not use TOP). 10
11. Write a query using an INNER JOIN to fetch the FullName and DeptName of all employees whose salary is greater than 50,000. 10
12. Write a query to find duplicate DeptID entries in the Employee table, displaying only those DeptIDs where the count of employees is greater than 5. 10
13. Create a Temporary Table named #TempEmp with two columns (ID, Name). Insert two dummy records into it. 10

SECTION D

(Case Study / Advanced Logic - 20 Marks)

Case Study: E-Commerce Orders

You are managing the backend database for an E-commerce application. You have access to the following table:

Orders (OrderID INT PRIMARY KEY, CustomerID INT, OrderDate DATE, TotalAmount FLOAT)

14. Read the above Case Study and answer the following:

- (a) Write a query to calculate the total sales (Sum of TotalAmount) generated by each CustomerID. (5 Marks) 5
- (b) Extend the query from part (a) to only show customers whose total aggregate sales exceed 100,000. (5 Marks) 5
- (c) Write the SQL syntax to create a User Defined Function (UDF) named GetDiscount. It should accept an Amount (FLOAT) as an input parameter and return a 10% discount value. (10 Marks) 10

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